

JEE MAIN REVISION MASTER: MAGNETISM AND MATTER

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Bar Magnet, Earth's Magnetism & Magnetic Materials	18 PYQs	Magnetism Properties

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Bar Magnet, Earth's Magnetism & Magnetic Materials	$\tan\theta = \frac{B_v}{B_h}$, $B = \sqrt{B_h^2 + B_v^2}$, $\mu_r = 1 + \chi_m$, $\text{Curie: } \chi = \frac{C}{T}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Bar Magnet, Earth's Magnetism & Magnetic Materials (Very High Yield — 18 PYQs)

Core Principle: Earth's magnetic elements: Declination, Dip angle θ where $\tan\theta = (B_v / B_h)$, and Horizontal component $B_h = B\cos\theta$. At the magnetic poles: $\theta = 90^\circ$, $B_h = 0$. At the magnetic equator: $\theta = 0^\circ$, $B_v = 0$. Material classification: Diamagnetic ($\chi < 0$, independent of T), Paramagnetic ($\chi > 0$, $\chi \propto 1/T$ Curie's law), Ferromagnetic ($\chi \gg 1$, loses ferromagnetism above Curie temperature T_c).

■ **Caution / JEE Pitfall:** Superconductors exhibit perfect diamagnetism with $\chi = -1$ and $\mu_r = 0$ (Meissner effect).

■ **Speed Trick:** If a bar magnet of moment M is cut into two equal halves along its length, each half has moment M/2. If cut perpendicular to its length, each half also has moment M/2.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substa..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If a paramagnetic bar is brought near a bar magnet, then it is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following statements is true in respect of diamagnetic substances?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substances with $-1 \leq \chi < 0$ are diamagnetic. II. Substances with $\chi \ll 1$ are paramagnetic. III. Substances with $\chi \gg 1$ are ferromagnetic. IV. Substances with $\mu_r \gg 1$ are ferromagnetic.

- (A) I and II
- (B) III and IV
- (C) II and III
- (D) II and IV

Correct Answer: None

Solution Analysis: The wrong statements are II and III. For magnetic substances, remember these NCERT-based facts: Diamagnetic substances have small negative susceptibility: $-1 \leq \chi < 0$. So statement I is correct. Paramagnetic substances have small positive susceptibility: $0 < \chi \ll 1$. They do not have $\chi \gg 1$. So statement II is wrong. Ferromagnetic substances have very large positive susceptibility: $\chi \gg 1$. They do not have $\chi \ll 1$. So statement III is wrong. For ferromagnetic substances, relative permeability is also very large: $\mu_r \gg 1$. So statement IV is correct. Hence, the wrong statements are: \$